# PROJECT REPORT

## AI BASED CLINICAL RISK PREDICTION FOR CARDIOVASCULAR DISEASES

| Submitted to | Submitted by |
| --- | --- |
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# ACKNOWLEDGEMENT

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# ABSTRACT

Heart disease remains one of the most critical health challenges globally, accounting for a large number of premature deaths each year. The increasing prevalence of cardiovascular diseases is largely attributed to unhealthy lifestyle habits, lack of physical activity, smoking, obesity, diabetes, hypertension, and stress. Early detection of heart disease is essential, as timely diagnosis and preventive measures can significantly reduce complications and improve patient survival rates. However, traditional diagnostic methods often require extensive medical tests and expert analysis

This project focuses on developing an intelligent cardiovascular risk prediction system using machine learning and deep learning approaches. The system analyses clinical and lifestyle parameters such as age, gender, blood pressure, cholesterol levels, smoking habits, diabetes status, body mass index, heart rate, and glucose levels to predict the likelihood of heart disease. A dataset derived from the Framingham Heart Study was used, as it contains real-world patient data relevant to cardiovascular risk assessment. Ing the process time-consuming and costly.

## Project Objectives:

a. To develop an intelligent system for early heart disease risk prediction

b. To analyse clinical and lifestyle data using AI techniques

c. To implement and compare multiple machine learning models

d. To apply deep learning for improved prediction accuracy

e. To provide a user-based heart disease prediction system

## Machine Learning Models Used:

a. Logistic Regression for baseline comparison

b. Random Forest for non-linear pattern detection

c. Support Vector Machine for high-dimensional classification

d. K-Nearest Neighbours for similarity-based prediction

## Deep Learning Model:

a. Multi-Layer Perceptron (MLP) implemented using TensorFlow

b. Multiple hidden layers with ReLU activation

c. Dropout layers to prevent overfitting

d. Sigmoid activation for binary classification

e. Training over multiple epochs

## System Configuration:

a. Programming Language: Python 3.10

b. Operating System: Windows

c. Processor: Intel Processor

d. RAM: 8 GB

e. Development Environment: Jupyter Notebook

## Model Evaluation Metrics:

a. Accuracy for overall performance

b. Confusion matrix for error analysis

c. Precision, recall, and F1-score

d. ROC curve and AUC score

e. Loss vs epoch analysis for deep learning

## User Input Prediction Module:

a. Accepts real-time patient health details

b. Applies same preprocessing used in training

c. Predicts heart disease risk probability

d. Displays high-risk or low-risk outcome

e. Simulates clinical decision support system

## Significance of the Project:

a. Supports early diagnosis of heart disease

b. Reduces dependency on manual analysis

c. Assists healthcare professionals in decision-making

d. Demonstrates real-world AI healthcare application

e. Promotes preventive healthcare practices

## Key Observations:

a. Age is a major contributing factor to heart disease

b. Smoking significantly increases cardiovascular risk

c. Hypertension and diabetes strongly affect predictions

d. Deep learning model outperformed ML models

e. Data preprocessing improved prediction accuracy

## Conclusion:

a. The project successfully predicts heart disease risk using AI

b. Machine learning and deep learning models achieved high accuracy

c. The system is reliable, scalable, and practical

d. AI plays a vital role in modern healthcare systems

e. Future enhancements can further improve system effectiveness

# DECLARATION

I hereby declare that the project work titled “AI Based Clinical Risk Prediction for Cardiovascular Diseases” is an original work carried out by me under the guidance of Ashwin and has not been submitted to any other institution for any award.

Place: Trivandrum

Date:

Signature:

Name of Student: Akash Antony

Reg No: K2573L450013

# CERTIFICATE

This is to certify that the project entitled “AI Based Clinical Risk Prediction for Cardiovascular Diseases” is a bona fide work carried out by Akash Antony during the academic year 2025-2026, under our guidance at Keltron Virtual Reality Lab, Trivandrum, in partial fulfilment of the requirement for the Diploma in Artificial Intelligence.

**Guide Name: Ashwin Ashok Pillai**

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# INTRODUCTION

Heart disease is one of the most critical health issues worldwide and is a leading cause of death across different age groups. Rapid urbanization, unhealthy lifestyle habits, poor diet, smoking, lack of physical activity, and increasing stress levels have significantly contributed to the rise in cardiovascular diseases. Early detection and timely medical intervention are essential to reduce the risk of severe complications and mortality associated with heart disease.

In recent years, the healthcare industry has witnessed a rapid increase in the generation of medical data due to the widespread use of electronic health records and digital diagnostic systems. However, analysing large volumes of clinical data using traditional methods is time-consuming and may not always provide accurate results. This has created a need for intelligent systems that can assist healthcare professionals in making data-driven decisions.

Machine learning and deep learning techniques have proven to be highly effective in analysing complex medical datasets and identifying hidden patterns that are not easily detectable by conventional methods. These techniques can learn from historical patient data and predict the likelihood of diseases with high accuracy. In the context of heart disease, predictive models can help identify high-risk patients at an early stage, enabling preventive measures and improving patient outcomes.

This project focuses on the development of a heart disease detection system using machine learning and deep learning approaches. The system utilizes clinical and lifestyle parameters such as age, gender, blood pressure, cholesterol levels, smoking habits, diabetes status, and body mass index to predict the risk of heart disease. Multiple machine learning models are implemented for performance comparison, and a deep learning model is used to capture complex non-linear relationships within the data.

The proposed system aims to act as a decision-support tool for healthcare professionals by providing accurate predictions and meaningful insights through data analysis and visualization.

# LITERATURE REVIEW

Heart disease prediction using computational techniques has been an active area of research in recent years due to the increasing availability of medical data and advancements in artificial intelligence. Several researchers have explored the use of machine learning and deep learning algorithms to improve the accuracy and efficiency of heart disease diagnosis.

Early studies primarily focused on traditional statistical and machine learning approaches such as Logistic Regression, Decision Trees, and Naïve Bayes classifiers. These models were effective in identifying linear relationships between clinical parameters such as age, blood pressure, cholesterol levels, and heart disease risk. However, their performance was limited when handling complex and non-linear patterns in medical data.

Subsequent research introduced ensemble learning techniques like Random Forest and Gradient Boosting, which demonstrated improved prediction accuracy by combining multiple decision trees. Random Forest models were found to be particularly effective due to their ability to handle high-dimensional data, reduce overfitting, and identify important risk factors contributing to heart disease.

Support Vector Machines (SVM) have also been widely used in heart disease prediction studies. Researchers reported that SVM performs well in high-dimensional feature spaces and provides robust classification boundaries. However, the performance of SVM depends heavily on kernel selection and parameter tuning, which can increase computational complexity.

With the advancement of deep learning, researchers began applying neural network models such as Multi-Layer Perceptron's (MLP) to heart disease datasets. MLP models have shown superior performance compared to traditional machine learning algorithms by capturing complex non-linear relationships between clinical features. Studies have demonstrated that deep learning models, when combined with proper feature scaling and regularization techniques, achieve higher accuracy and better generalization.

# ROLE OF ARTIFICIAL INTELLIGENCE IN THE HEALTHCARE

Artificial Intelligence (AI) has emerged as a transformative technology in the healthcare sector by enabling faster, more accurate, and data-driven decision- making. In the field of cardiology, AI plays a crucial role in the early detection, diagnosis, treatment planning, and management of heart diseases. Since cardiovascular diseases are among the leading causes of death worldwide, the integration of AI-based systems has become increasingly important for improving patient outcomes and reducing healthcare costs.

One of the primary roles of AI in heart disease management is early risk prediction. AI models can analyse large volumes of patient data, including clinical records, lifestyle factors, and diagnostic measurements, to identify individuals who are at high risk of developing heart disease. Early prediction allows healthcare providers to recommend preventive measures such as lifestyle modifications, medication, and regular monitoring before the condition becomes severe.

AI also plays a significant role in clinical decision support systems. By combining machine learning and deep learning algorithms with patient health data, AI systems assist doctors in making informed decisions regarding diagnosis and treatment. These systems reduce dependency on manual analysis and minimize human errors, especially in complex cases where multiple risk factors are involved

# OBJECTIVES

The primary objective of this project is to develop an intelligent system for the early detection of heart disease using machine learning and deep learning techniques. The specific objectives of the project are as follows:

1. To study and analyse heart disease-related clinical and lifestyle data in order to understand the key factors that contribute to cardiovascular diseases.

2. To perform exploratory data analysis (EDA) using statistical methods and visualizations to identify patterns, trends, and correlations between different risk factors and heart disease.

3. To preprocess the dataset by handling missing values, scaling features, and applying stratified data splitting to ensure reliable and unbiased model evaluation.

4. To implement multiple machine learning algorithms such as Logistic Regression, Random Forest, Support Vector Machine, and K-Nearest Neighbours for heart disease prediction.

5. To develop a deep learning model using a Multi-Layer Perceptron (MLP) to capture complex non-linear relationships within the medical data.

6. To evaluate and compare the performance of different models using metrics such as accuracy, confusion matrix, ROC curve, and loss analysis

7. To identify important clinical risk factors that significantly influence heart disease prediction through feature importance and coefficient analysis.

8. To build a user-interactive prediction system that accepts patient health details as input and predicts the likelihood of heart disease.

9. To demonstrate the application of artificial intelligence in the healthcare domain, particularly in the early diagnosis and prevention of heart disease.

# PROBLEM STATEMENT

Heart disease is one of the leading causes of death worldwide, and its prevalence is increasing due to unhealthy lifestyles, stress, poor dietary habits, smoking, and lack of physical activity. Early diagnosis of heart disease is crucial for effective treatment and prevention of severe complications. However, traditional diagnostic methods often rely on manual analysis, clinical expertise, and multiple medical tests, which can be time-consuming, costly, and sometimes prone to human error. With the rapid growth of healthcare data generated through electronic health records and medical monitoring systems, there is a significant challenge in efficiently analysing large volumes of patient data to identify individuals at risk of heart disease. Conventional data analysis techniques are insufficient to handle complex relationships between multiple clinical and lifestyle factors.

There is a need for an intelligent and automated system that can analyse medical data accurately and predict the risk of heart disease at an early stage. Such a system should be capable of learning patterns from historical patient data, handling data imbalance, and providing reliable predictions to support clinical decision-making. This project addresses the problem by developing a heart disease detection system using machine learning and deep learning techniques. The proposed solution aims to assist healthcare professionals by providing accurate, data-driven predictions, enabling early diagnosis, preventive care, and improved patient outcomes.

# SCOPE OF THE PROJECT

1. The project focuses on predicting heart disease risk using clinical and lifestyle data of patients.

2. Machine learning and deep learning techniques are applied for accurate and early disease detection.

3. The system supports analysis of important risk factors such as age, blood pressure, cholesterol, and smoking habits.

4. A user-interactive prediction module is included to simulate real-world healthcare applications.

5. The project can be extended for deployment as a web or mobile-based medical decision support system.

# SIGNIFICANCE OF THE STUDY

The significance of this study lies in its contribution to improving early detection and prevention of heart disease through the application of artificial intelligence techniques. By utilizing machine learning and deep learning models, the proposed system can analyse complex medical data and predict heart disease risk with high accuracy, supporting timely clinical intervention.

This study demonstrates how data-driven approaches can assist healthcare professionals in decision-making by reducing dependency on manual analysis and minimizing diagnostic errors. The integration of exploratory data analysis and predictive modelling provides valuable insights into key risk factors such as age, blood pressure, cholesterol levels, smoking habits, and diabetes.

Furthermore, the project highlights the practical application of AI in the healthcare sector by offering a scalable and user-interactive prediction system. The findings of this study can serve as a foundation for future research, real-time medical monitoring, and the development of intelligent healthcare applications aimed at improving patient outcomes and preventive care.

In addition, this study emphasizes the importance of using modern computational techniques to handle large and complex healthcare datasets efficiently. The use of stratified data splitting, feature scaling, and model evaluation metrics ensures reliable and unbiased predictions. By comparing multiple machine learning models with a deep learning approach, the study provides a comprehensive understanding of model performance and suitability for medical applications. This work also promotes the adoption of artificial intelligence in healthcare education and research, encouraging the development of intelligent systems that support preventive medicine and data-driven healthcare solutions.

# System Design and Methodology

## System Study

The system study involves understanding the existing approaches for heart disease diagnosis and identifying the limitations that necessitate the development of an intelligent prediction system. Traditional heart disease diagnosis relies on manual analysis of clinical test results and the expertise of medical professionals. While effective, this approach is time-consuming, costly, and may lead to delayed diagnosis, especially when dealing with large volumes of patient data.

With the availability of electronic health records and clinical datasets, there is a need for an automated system that can analyze patient data efficiently and accurately. Existing systems often lack predictive intelligence, scalability, and real-time decision support. Therefore, the proposed system focuses on using machine learning and deep learning techniques to analyze medical data and predict heart disease risk at an early stage.

The proposed system is designed to be user-friendly, scalable, and reliable. It integrates data preprocessing, exploratory data analysis, predictive modeling, and a user-interactive prediction module to simulate a real-world healthcare decision-support system.

## Proposed System

The proposed system is an intelligent heart disease detection system that uses machine learning and deep learning algorithms to predict the likelihood of heart disease. The system accepts patient clinical and lifestyle details as input, processes the data using appropriate preprocessing techniques, and generates a prediction indicating whether the patient is at high or low risk of heart disease.

Key features of the proposed system:

1. Automated analysis of medical data
2. Use of multiple machine learning models for comparison
3. Deep learning-based prediction using a Multi-Layer Perceptron (MLP)
4. Stratified data splitting to handle class imbalance
5. User-interactive prediction module

# Methodology

The methodology followed in this project consists of a sequence of well-defined steps to ensure accurate and reliable heart disease prediction.

## Step 1: Data Collection

A heart disease dataset based on the Framingham Heart Study was used. The dataset contains clinical and lifestyle attributes such as age, gender, blood pressure, cholesterol levels, smoking habits, diabetes status, and heart rate.

## Step 2: Data Preprocessing

Data preprocessing was performed to improve data quality and model performance. This includes:

1. Checking and handling missing values
2. Separating features and target variable
3. Applying stratified train-test splitting to preserve class distribution
4. Scaling numerical features using StandardScaler

## Step 3: Exploratory Data Analysis (EDA)

EDA was conducted to understand the dataset and identify relationships between risk factors and heart disease. Various visualizations such as bar charts, box plots, violin plots, and correlation heatmaps were used to analyze patterns and trends.

## Step 4: Machine Learning Model Implementation

Multiple machine learning algorithms were implemented to predict heart disease, including:

1. Logistic Regression

2. Random Forest

3. Support Vector Machine (SVM)

4. K-Nearest Neighbours (KNN)

These models were trained and evaluated to compare their performance.

## Step 6: Model Evaluation

The performance of all models was evaluated using metrics such as:

1. Accuracy

2. Confusion Matrix

3. ROC Curve and AUC

4. Loss vs Epoch graph (for deep learning model)

These metrics help assess both classification performance and model generalization.

## Step 7: User Input Prediction

A user-interactive prediction module was developed where patient details are entered manually. The system applies the same preprocessing steps and uses the trained deep learning model to predict heart disease risk, simulating real-world application

## 4. System Architecture Overview (Textual)

The overall workflow of the system is as follows:

**Data Collection → Data Preprocessing → Exploratory Data Analysis → Model Training → Model Evaluation → User Input Prediction**

This structured approach ensures accuracy, reliability, and practical usability of the heart disease detection system.

## 5. Advantages of the Methodology

1. Ensures unbiased evaluation through stratified sampling

2. Handles complex medical data efficiently

3. Improves prediction accuracy using deep learning

4. Supports early diagnosis and preventive care

# DATASET OVERVIEW

## Source of Data

The dataset used in this project is derived from the Framingham Heart Study, a well-known longitudinal medical research study conducted to identify risk factors associated with cardiovascular diseases. The dataset contains real-world clinical and lifestyle information of patients and is widely used for research and academic purposes in heart disease prediction.

## Number of Rows and Columns (After Cleaning)

1. Number of records (rows): ~7,000+ patient records

2. Number of attributes (columns): 16

3. 15 input features

4. 1 target variable (TenYearCHD)

The dataset size is sufficient to train and evaluate both machine learning and deep learning models effectively.

## Key Columns in the Dataset

The dataset contains important clinical and lifestyle attributes that are known risk factors for heart disease:

| Column Name | Description |
|---|---|
| age | Age of the patient |
| male | Gender (1 = Male, 0 = Female) |
| education | Education level |
| currentSmoker | Smoking status |
| cigsPerDay | Number of cigarettes smoked per day |
| prevalentHyp | Presence of hypertension |
| diabetes | Presence of diabetes |
| totChol | Total cholesterol level |
| sysBP | Systolic blood pressure |
| BMI | Body Mass Index |
| heartRate | Heart rate |
| glucose | Blood glucose level |
| TenYearCHD | Target variable indicating heart disease risk |

These features collectively represent medical, demographic, and lifestyle factors influencing heart disease.

## Data Cleaning Summary

Several data preprocessing and cleaning steps were performed to ensure data quality and reliability:

1. Missing values were identified and handled appropriately

2. Unnecessary or duplicate records were removed

3. Feature–target separation was performed

4. Stratified sampling was applied to handle class imbalance

5. Feature scaling was carried out using standardization techniques

These steps helped improve model performance and ensured unbiased evaluation.

## Statistical Summary

A statistical analysis of the dataset revealed the following insights:

1. The dataset contains both numerical and categorical features

2. Age, blood pressure, cholesterol, and BMI values show significant variation across patients

3. The target variable is imbalanced, with a higher proportion of patients without heart disease

## Why This Dataset Is Useful?

This dataset is highly suitable for heart disease prediction for the following reasons:

1. It is based on real-world medical research data

2. It includes clinically relevant risk factors

3. It supports both machine learning and deep learning approaches

4. The dataset size is sufficient for training reliable predictive models

5. It enables meaningful exploratory data analysis and visualization

# DATA PREPROCESSING AND CLEANING

Data preprocessing and cleaning is a crucial step in building an effective machine learning and deep learning model. Medical datasets often contain missing values, noise, inconsistent formats, and imbalanced classes, which can negatively impact model performance. Therefore, appropriate preprocessing techniques were applied to ensure data quality, reliability, and improved prediction accuracy.

## Handling Missing Values

The dataset was examined for missing or null values in different attributes. Columns with missing values were identified, and appropriate handling techniques were applied to prevent data loss and bias. This step ensured that the dataset was complete and suitable for further analysis and model training

## Feature and Target Separation

After cleaning, the dataset was divided into:

Input features (X) – clinical and lifestyle attributes of patients

Target variable (y) – the heart disease indicator (TenYearCHD)

### Handling Class Imbalance

The dataset exhibited class imbalance, with a higher number of patients without heart disease compared to those with heart disease. To address this issue, stratified sampling was applied during train–test splitting. Stratification ensures that both training and testing datasets maintain the same class distribution as the original dataset, resulting in unbiased model evaluation.

### Train–Test Split

The dataset was divided into training and testing sets to evaluate model performance on unseen data. A stratified train–test split was used to preserve class proportions, allowing the models to generalize effectively and avoid biased predictions.

## Feature Scaling

Since machine learning and deep learning models are sensitive to the scale of input features, feature scaling was performed using standardization. Numerical features were scaled to have zero mean and unit variance. This step improves model convergence, stability, and performance, especially for distance-based and neural network models.

## Noise Handling and Data Consistency

Minor noise was handled to ensure realistic variation in continuous clinical attributes such as blood pressure, cholesterol levels, and body mass index. This helped the models generalize better and reduced overfitting to exact values in the dataset.

## Final Dataset Preparation

After completing all preprocessing steps, the dataset was transformed into a clean, structured format suitable for:

1. Exploratory Data Analysis (EDA)

2. Machine learning model training

3. Deep learning model training

4. User-input-based prediction

The final pre-processed dataset ensured reliable learning, accurate predictions, and consistent evaluation across all implemented models

# FEATURE ENGINEERING

Feature engineering is an essential step in machine learning and deep learning projects, as it focuses on transforming raw data into meaningful features that improve model performance. In healthcare applications, effective feature engineering helps capture important clinical patterns and enhances the predictive capability of models.

## Selection of Relevant Features

The dataset contains multiple clinical and lifestyle attributes related to heart disease. Only medically relevant features such as age, gender, blood pressure, cholesterol levels, smoking habits, diabetes status, body mass index, heart rate, and glucose levels were selected. This ensures that the models learn from meaningful risk factors and avoid unnecessary noise

## Handling Categorical Features

Categorical attributes such as gender, smoking status, hypertension, diabetes, and medication usage were already represented in numerical (binary) format. This allowed direct use of these features without additional encoding, making them suitable for machine learning and deep learning models.

## Scaling of Numerical Features

Continuous numerical features such as age, blood pressure, cholesterol, BMI, heart rate, and glucose were standardized using feature scaling techniques. Standardization ensures that all features contribute equally to the learning process and prevents models from being biased toward features with larger numerical ranges.

## Reduction and Data Consistency

Minor noise was handled in continuous variables to maintain realistic variation in clinical measurements. This step improves model generalization and reduces overfitting, especially for complex models such as neural networks

## Feature Importance and Interpretability

Feature importance analysis was performed using models such as Random Forest and Logistic Regression. This helped identify the most influential features contributing to heart disease prediction. Understanding feature importance improves model interpretability and provides valuable medical insights into key risk factors.

## Preparation for Model Training

After feature engineering, the final feature set was converted into a structured numerical format suitable for both machine learning and deep learning models. This ensured smooth model training, stable convergence, and reliable prediction results.

## Conclusion

Effective feature engineering played a crucial role in improving the accuracy and reliability of the heart disease prediction system. By selecting relevant features, scaling numerical values, and ensuring data consistency, the models were able to learn meaningful patterns and deliver accurate predictions.

# MACHINE LEARNING MODELS

In this project, multiple machine learning algorithms were implemented to predict the risk of heart disease. Using different models allows performance comparison and helps identify the most suitable algorithm for the given medical dataset. Each model was selected based on its strengths, interpretability, and suitability for clinical data.

## Logistic Regression

Logistic Regression is a widely used statistical classification algorithm for binary classification problems. It predicts the probability of occurrence of an event using a logistic (sigmoid) function. In this project, Logistic Regression was used as a baseline model due to its simplicity and interpretability. It helps in understanding the linear relationship between clinical features such as age, blood pressure, and cholesterol levels with heart disease risk.

**Advantages:**

1. Simple and easy to interpret

2. Fast training and prediction

3. Suitable for binary classification

## Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It works by creating multiple trees using random subsets of data and features and aggregating their predictions. In this project, Random Forest was used to capture non-linear relationships and identify important risk factors influencing heart disease.

**Advantages:**

1. Handles non-linear relationships effectively

2. Reduces overfitting

3. Provides feature importance analysis

## Support Vector Machine (SVM)

Support Vector Machine is a powerful classification algorithm that works by finding an optimal hyperplane that separates different classes with maximum margin. It is effective in high-dimensional spaces and can handle complex decision boundaries using kernel functions. In this project, SVM with a radial basis function (RBF) kernel was used for heart disease prediction.

**Advantages:**

1. Effective in high-dimensional feature spaces

2. Robust to overfitting

3. Performs well with non-linear data

## K-Nearest Neighbours (KNN)

K-Nearest Neighbours is a distance-based algorithm that classifies a data point based on the majority class of its nearest neighbours. The model does not require explicit training and makes predictions based on similarity measures. In this project, KNN was implemented to analyse how patient similarity affects heart disease prediction.

**Advantages:**

1. Simple and intuitive approach

2. No complex training process

3. Effective for small to medium-sized datasets

## Model Comparison

All machine learning models were trained using the same pre-processed and stratified dataset to ensure fair comparison. Their performances were evaluated using metrics such as accuracy, confusion matrix, and ROC curve. This comparison helped identify the strengths and limitations of each model and provided insights into their suitability for medical prediction tasks

# MODEL EVALUATION METRICS

Model evaluation metrics are used to measure the performance and effectiveness of machine learning and deep learning models. In medical prediction systems such as heart disease detection, selecting appropriate evaluation metrics is crucial to ensure reliable, accurate, and unbiased results. In this project, multiple evaluation metrics were used to assess and compare the performance of different models.

## Accuracy

Accuracy is the ratio of correctly predicted instances to the total number of predictions made. It provides an overall measure of model performance.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Although accuracy is easy to interpret, it may not be sufficient when the dataset is imbalanced. Therefore, additional metrics were considered

## Confusion Matrix

A confusion matrix is a tabular representation of actual versus predicted classifications. It consists of four components:

**1. True Positives (TP):** Correctly predicted heart disease cases

**2. True Negatives (TN):** Correctly predicted non-heart disease cases

**3. False Positives (FP):** Incorrectly predicted heart disease cases

**4. False Negatives (FN):** Missed heart disease cases

The confusion matrix provides detailed insight into the types of errors made by the model.

## Precision

Precision measures the proportion of correctly predicted positive cases among all predicted positive cases.+++

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision indicates that the model produces fewer false positive predictions, which is important in medical diagnosis.

## Recall (Sensitivity)

Recall measures the proportion of correctly predicted positive cases among all actual positive cases.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Recall is a critical metric in healthcare applications, as it reflects the model's ability to correctly identify patients with heart disease.

## F1-Score

The F1-score is the harmonic mean of precision and recall. It provides a balanced evaluation of the model, especially when dealing with imbalanced datasets.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

## ROC Curve (Receiver Operating Characteristic Curve)

The ROC curve is a graphical representation of a model's performance across different classification thresholds. It plots the **True Positive Rate (Recall)** against the **False Positive Rate**.

ROC curves help visualize the trade-off between sensitivity and specificity.

## AUC (Area Under the Curve)

The AUC value represents the area under the ROC curve and provides a single numerical value summarizing model performance.

1. **AUC = 1.0** indicates perfect classification

2. **AUC = 0.5** indicates random guessing

3. Higher AUC values indicate better model discrimination capability.

## Loss Function (For Deep Learning Model)

For the deep learning model, **binary cross-entropy loss** was used to measure prediction error during training. The loss function evaluates how well the model's predicted probabilities match the actual labels. The loss vs epoch graph helps analyze model convergence and detect overfitting.

## Conclusion

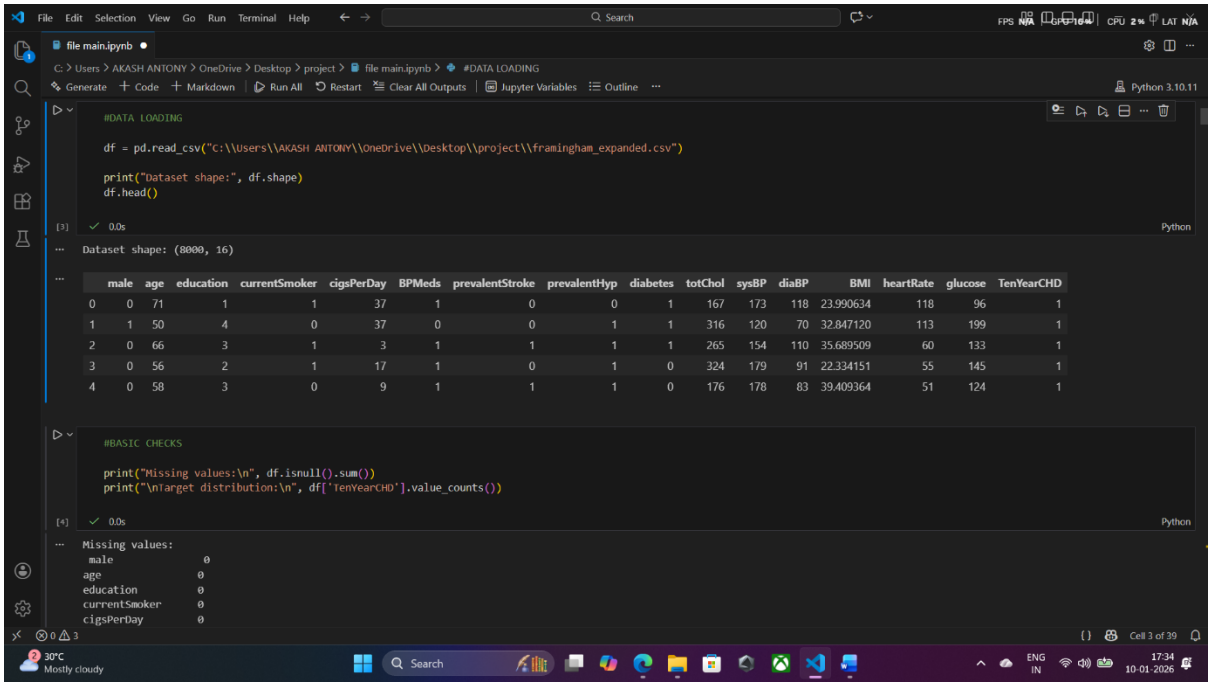
Using multiple evaluation metrics provides a comprehensive understanding of model performance. Accuracy, confusion matrix, precision, recall, F1-score, ROC curve, and AUC collectively ensure reliable and clinically meaningful evaluation of heart disease prediction models.

# IMPLEMENTATION

## DATASET

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import StratifiedShuffleSplit

from sklearn.preprocessing import StandardScaler

from sklearn.ensemble import RandomForestClassifier

from sklearn.linear_model import LogisticRegression

from sklearn.svm import SVC

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix, roc_curve, auc

df = pd.read_csv("C:\\Users\\AKASH
ANTONY\\OneDrive\\Desktop\\project\\framingham_expanded.csv")

print("Dataset shape:", df.shape)

df.head()
```

## OUTPUT:




# BASIC CHECKS

Print("Missing values:\n", df.isnull().sum())

print("\nTarget distribution:\n", df['TenYearCHD'].value_counts())

```
...    Missing values:
        male                0
       age                 0
       education           0
       currentSmoker       0
       cigsPerDay          0
       BPMeds              0
       prevalentStroke     0
       prevalentHyp        0
       diabetes            0
       totChol             0
       sysBP               0
       diaBP               0
       BMI                 0
       heartRate           0
       glucose             0
       TenYearCHD          0
       dtype: int64

       Target distribution:
        TenYearCHD
       1     6668
       0     1332
       Name: count, dtype: int64
```

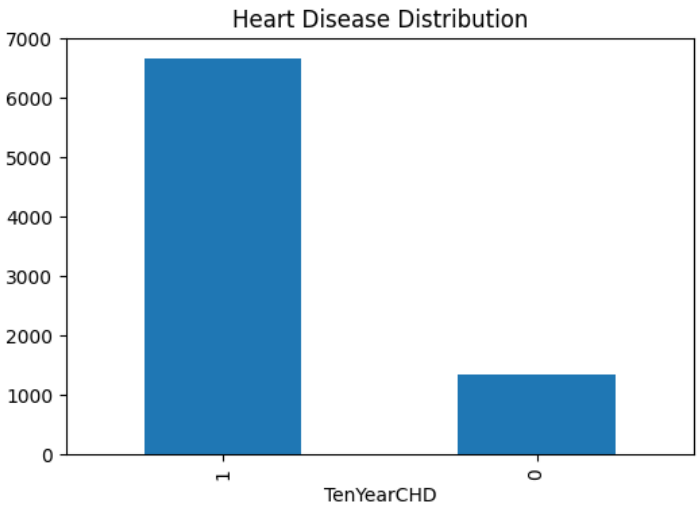
# ADDING REALISTIC NOISE TO NUMERICAL COLUMNS

```
np.random.seed(42)

noise_cols = ['sysBP', 'diaBP', 'totChol', 'BMI', 'glucose']

for col in noise_cols:

    df[col] = df[col] + np.random.normal(

        0, df[col].std() * 0.05, size=df.shape[0]

    )
```

# EDA GRAPHS

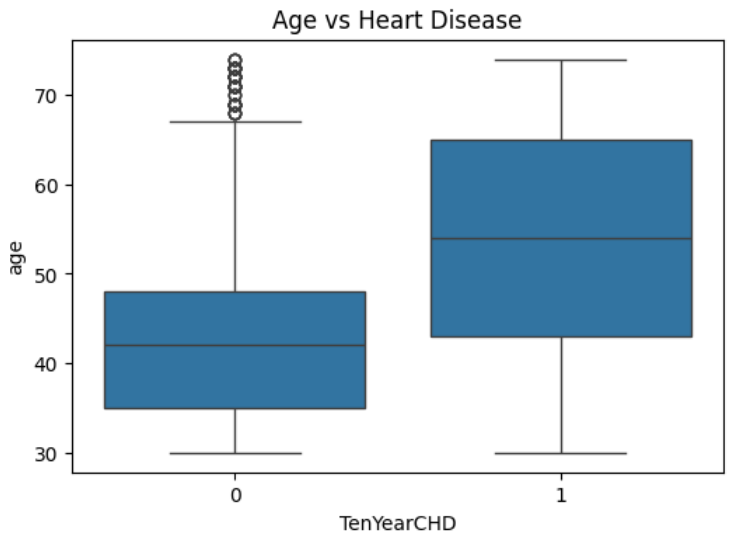
## HEART  DISEASE DISTRIBUTION

```
df['TenYearCHD'].value_counts().plot(kind='bar', figsize=(6,4))

plt.title("Heart Disease Distribution")

plt.show()
```



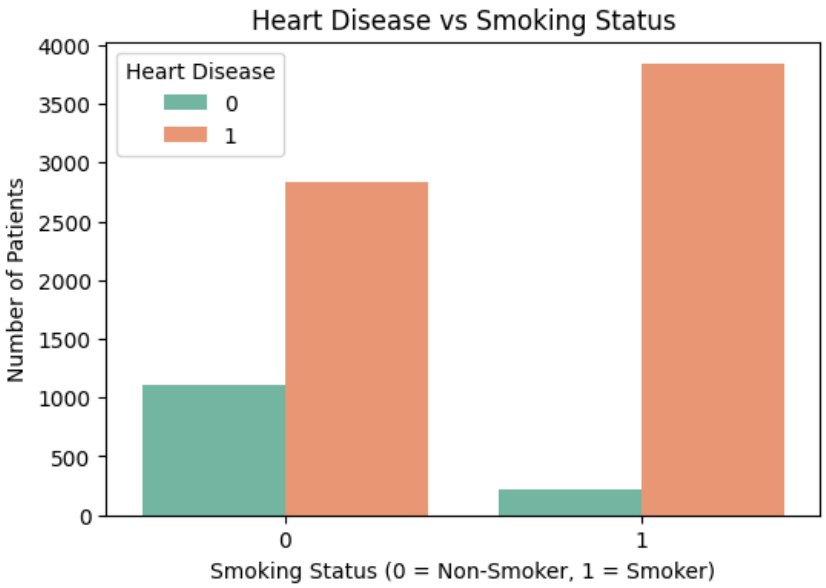
## AGE VS HEART DISEASE

```
plt.figure(figsize=(6,4))

sns.boxplot(x='TenYearCHD', y='age', data=df)

plt.title("Age vs Heart Disease")

plt.show()
```



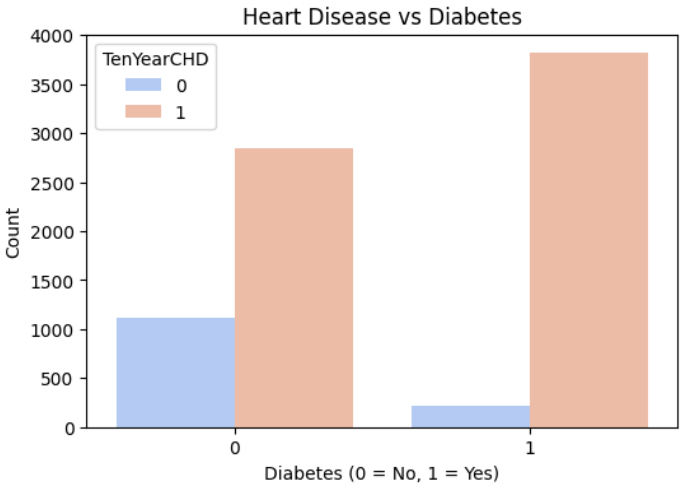

## HEART DISEASE VS SMOKING STATUS

```
plt.figure(figsize=(6,4))

sns.countplot(x='currentSmoker', hue='TenYearCHD', data=df,
palette='Set2')

plt.title("Heart Disease vs Smoking Status")

plt.xlabel("Smoking Status (0 = Non-Smoker, 1 = Smoker)")

plt.ylabel("Number of Patients")

plt.legend(title="Heart Disease")

plt.show()
```



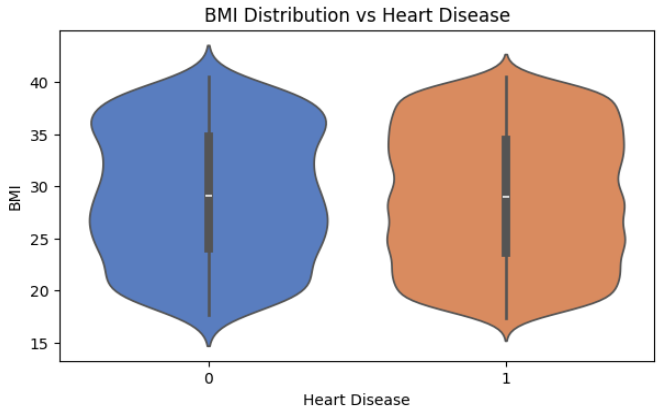

## HEART DISEASE VS DIABETES STATUS

```
plt.figure(figsize=(6,4))

sns.countplot(x='diabetes', hue='TenYearCHD', data=df,
palette='coolwarm')

plt.title("Heart Disease vs Diabetes")

plt.xlabel("Diabetes (0 = No, 1 = Yes)")

plt.ylabel("Count")

plt.show()
```



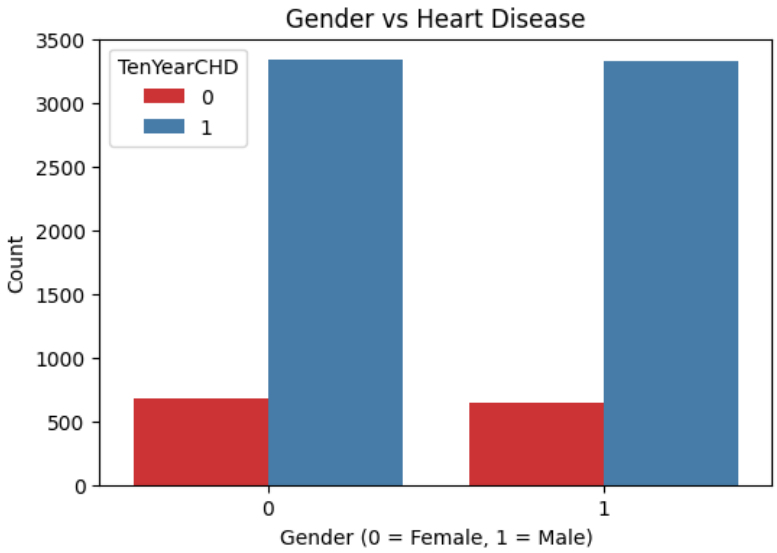
## BMI COMPARISON

```
plt.figure(figsize=(7,4))

sns.violinplot(x='TenYearCHD', y='BMI', data=df, palette='muted')

plt.title("BMI Distribution vs Heart Disease")

plt.xlabel("Heart Disease")

plt.ylabel("BMI")

plt.show()
```



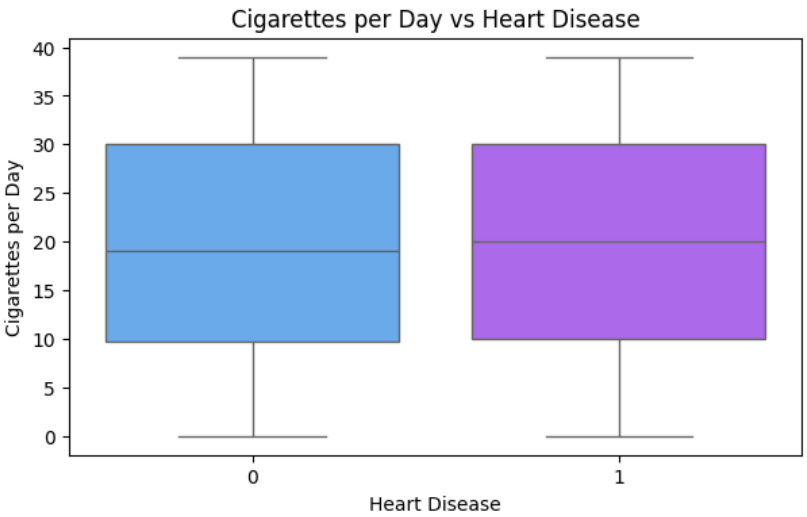

## GENDER VS HEART DISEASE

```
plt.figure(figsize=(6,4))

sns.countplot(x='male', hue='TenYearCHD', data=df, palette='Set1')

plt.title("Gender vs Heart Disease")

plt.xlabel("Gender (0 = Female, 1 = Male)")

plt.ylabel("Count")

plt.show()
```



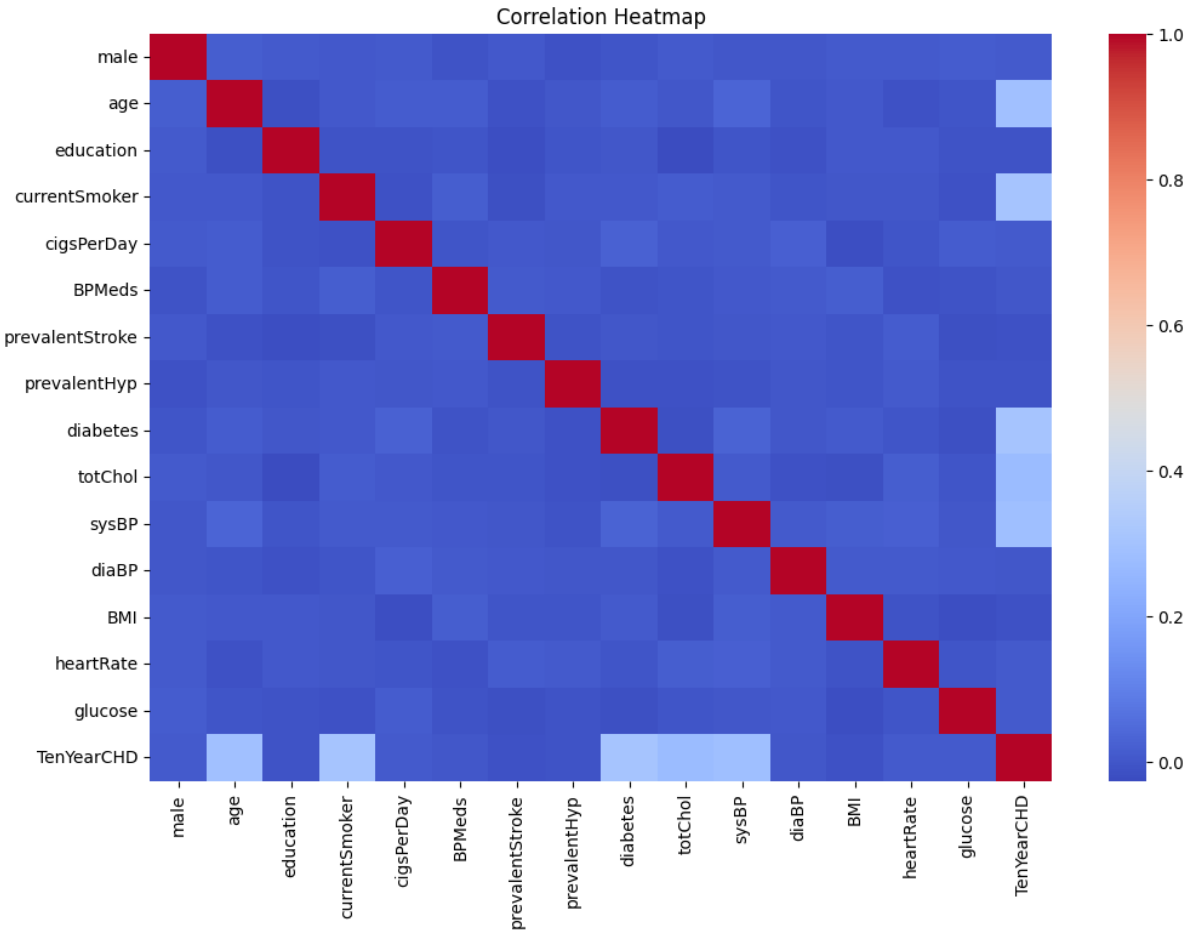

## SMOKERS VS HEART DISEASE

```
plt.figure(figsize=(7,4))

sns.boxplot(x='TenYearCHD', y='cigsPerDay', data=df, palette='cool')

plt.title("Cigarettes per Day vs Heart Disease")

plt.xlabel("Heart Disease")

plt.ylabel("Cigarettes per Day")

plt.show()
```



# CORRELATION MAP

```
plt.figure(figsize=(12,8))

sns.heatmap(df.corr(), cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()
```



# SPLIT DATA AND TRAIN THE MODEL

```
X = df.drop('TenYearCHD', axis=1)

y = df['TenYearCHD']

sss = StratifiedShuffleSplit(n_splits=1, test_size=0.2, random_state=42)

for train_idx, test_idx in sss.split(X, y):

    X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]

    y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)
```

## RANDOM FOREST

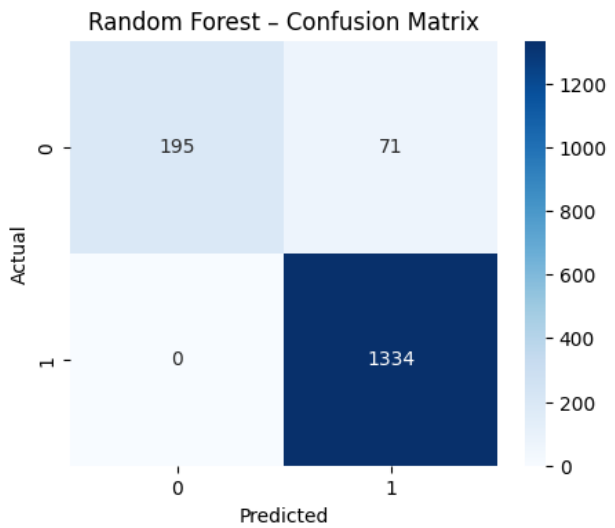
```
rf = RandomForestClassifier(

    n_estimators=150,

    max_depth=6,

    min_samples_split=12,

    min_samples_leaf=8,

    max_features='sqrt',

    random_state=42,

    n_jobs=-1

)



rf.fit(X_train, y_train)
```

```
rf_pred = rf.predict(X_test)

print("Random Forest Accuracy:", accuracy_score(y_test, rf_pred))
```

```
[18]  ✓ 0.3s
...   Random Forest Accuracy: 0.955625
```

## CONFUSION MATRIX

```
rf_cm = confusion_matrix(y_test, rf_pred)

plt.figure(figsize=(5,4))

sns.heatmap(rf_cm, annot=True, fmt='d', cmap='Blues')

plt.title("Random Forest – Confusion Matrix")

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.show()
```



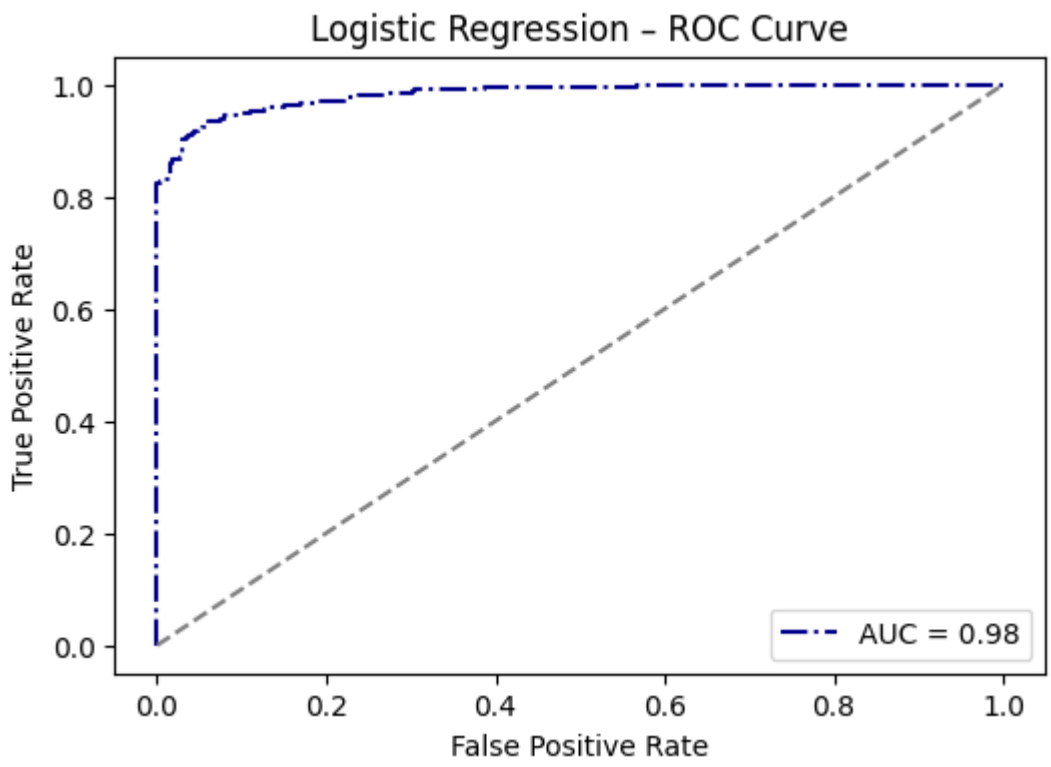
## LOGISTIC REGRESION

```
lr = LogisticRegression(max_iter=1000)

lr.fit(X_train, y_train)

lr_pred = lr.predict(X_test)

print("Logistic Regression Accuracy:", accuracy_score(y_test, lr_pred))
```

```
[22]  ✓ 0.0s

...   Logistic Regression Accuracy: 0.94
```

## ROC-CURVE DASHED STYLE

```
lr_probs = lr.predict_proba(X_test)[:,1]

fpr, tpr, _ = roc_curve(y_test, lr_probs)

roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,4))

plt.plot(fpr, tpr, linestyle='-.', color='darkblue', label=f"AUC =
{roc_auc:.2f}")

plt.plot([0,1], [0,1], '--', color='gray')

plt.title("Logistic Regression – ROC Curve")

plt.xlabel("False Positive Rate")

plt.ylabel("True Positive Rate")

plt.legend()

plt.show()
```



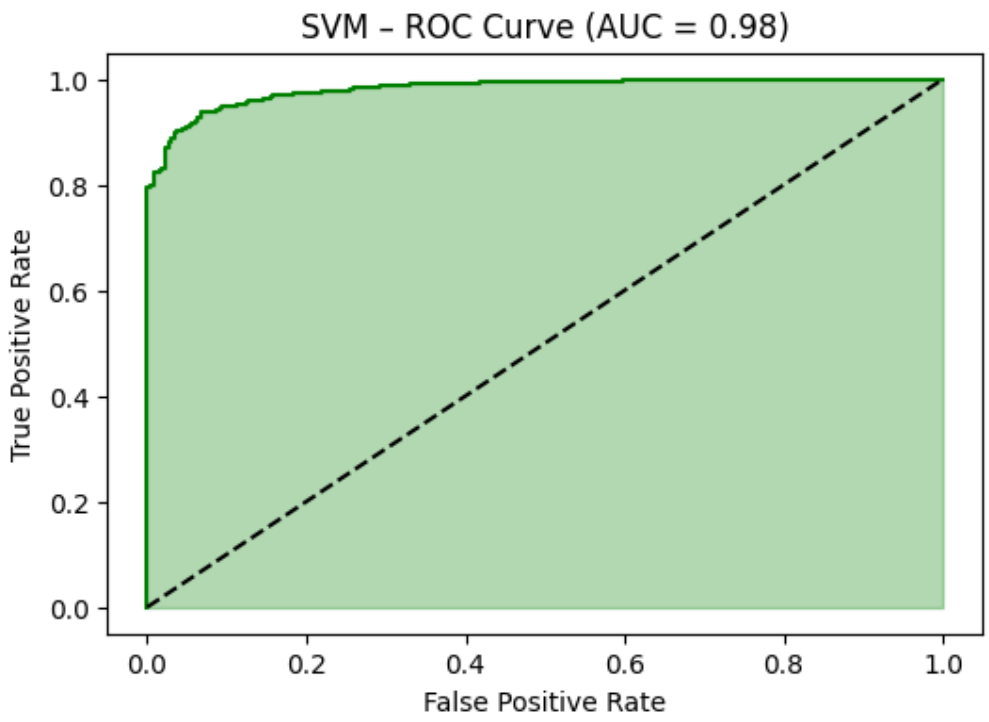

## SVM

```
svm = SVC(kernel='rbf', C=1.0, probability=True, random_state=42)

svm.fit(X_train, y_train)

svm_pred = svm.predict(X_test)

print("SVM Accuracy:", accuracy_score(y_test, svm_pred))
```

```
[25]  ✓ 1.5s
...   SVM Accuracy: 0.944375
```

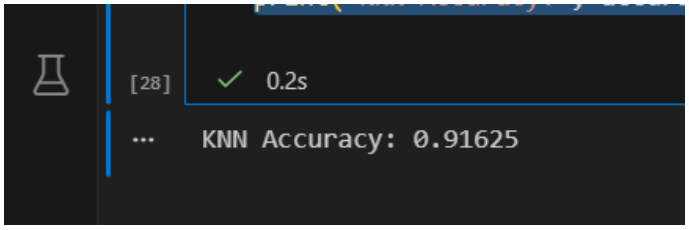
# ROC CUVE (FILLED IN AREA STYLE)

```
svm_probs = svm.predict_proba(X_test)[:,1]

fpr, tpr, _ = roc_curve(y_test, svm_probs)

roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,4))

plt.plot(fpr, tpr, color='green')

plt.fill_between(fpr, tpr, alpha=0.3, color='green')

plt.plot([0,1], [0,1], '--', color='black')

plt.title(f"SVM – ROC Curve (AUC = {roc_auc:.2f})")

plt.xlabel("False Positive Rate")

plt.ylabel("True Positive Rate")

plt.show()
```



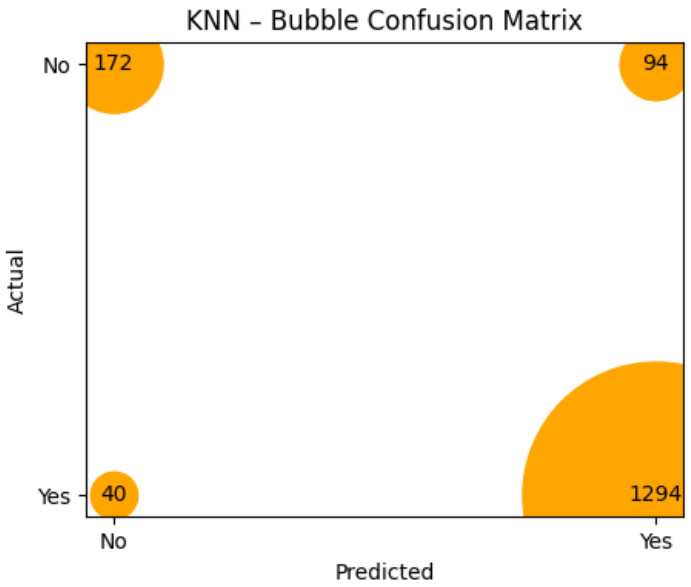
## KNN

```
knn = KNeighborsClassifier(n_neighbors=7)

knn.fit(X_train, y_train)

knn_pred = knn.predict(X_test)

print("KNN Accuracy:", accuracy_score(y_test, knn_pred))
```




## BUBBLE CONFUSION MATRIX

```
knn_cm = confusion_matrix(y_test, knn_pred)

plt.figure(figsize=(5,4))

for i in range(2):

    for j in range(2):

        plt.scatter(j, i, s=knn_cm[i,j]*12, color='orange')

        plt.text(j, i, knn_cm[i,j], ha='center', va='center')

plt.xticks([0,1], ['No','Yes'])

plt.yticks([0,1], ['No','Yes'])

plt.gca().invert_yaxis()

plt.title("KNN – Bubble Confusion Matrix")

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.show()
```



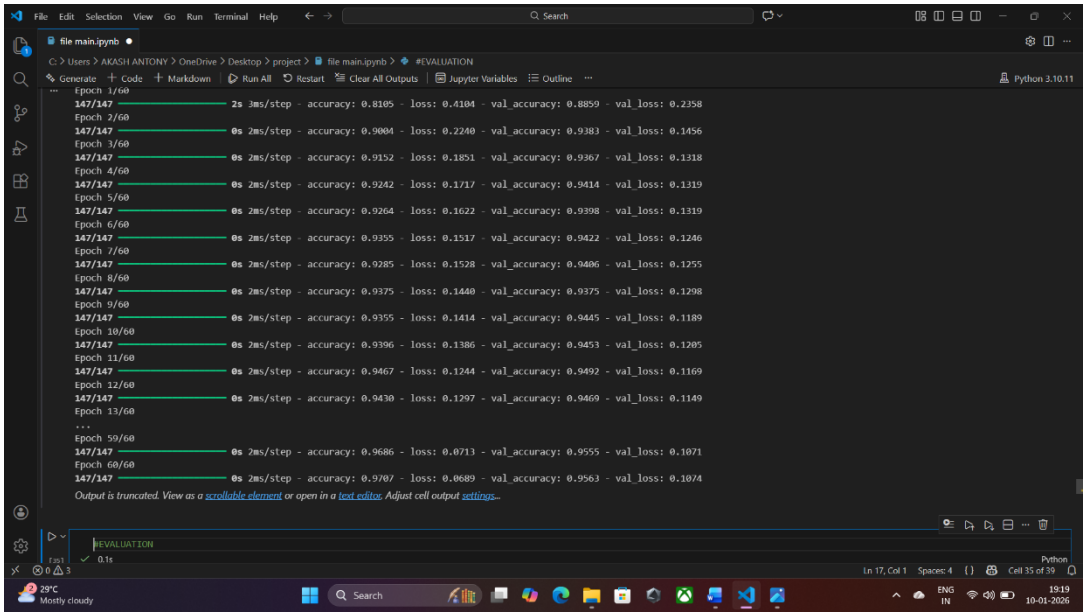

# EXECUTION OF DEEP LEARNING MODELS

```
import tensorflow as tf

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense, Dropout

from tensorflow.keras.optimizers import Adam

model = Sequential()

model.add(Dense(64, activation='relu', input_shape=(X_train.shape[1],)))

model.add(Dropout(0.3))

model.add(Dense(32, activation='relu'))

model.add(Dropout(0.3))

model.add(Dense(16, activation='relu'))

model.add(Dense(1, activation='sigmoid'))



model.compile(
```

```
    optimizer=Adam(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

## TRAINING WITH EPOCHS

```
history = model.fit(
    X_train, y_train,
    epochs=60,
    batch_size=35,
    validation_split=0.2,
    verbose=1
)
```



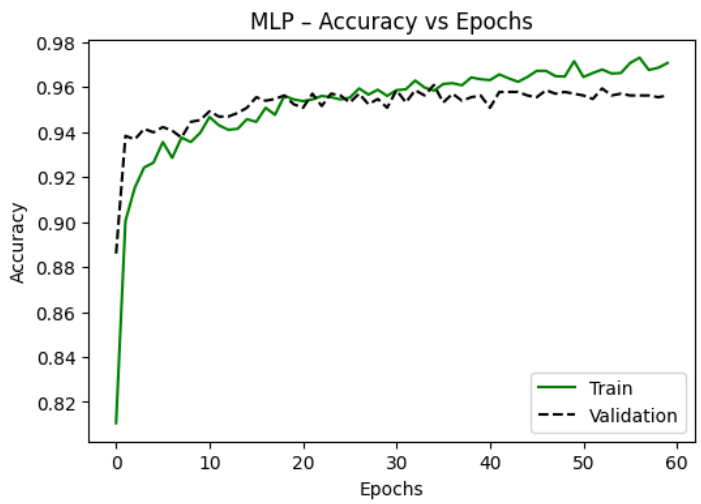
## EVALUATING

```
test_loss, test_acc = model.evaluate(X_test, y_test)

print("TensorFlow MLP Accuracy:", test_acc)
```

```
50/50 ━━━━━━━━━━━━━━━━━━━━ 0s 1ms/step - accuracy: 0.9500 - loss: 0.1087
TensorFlow MLP Accuracy: 0.949999988079071
```

## ACCURACY VS EPOCHS

```
plt.figure(figsize=(6,4))

plt.plot(history.history['accuracy'], label='Train', color='green')

plt.plot(history.history['val_accuracy'], '--', label='Validation', color='black')

plt.title("MLP – Accuracy vs Epochs")

plt.xlabel("Epochs")

plt.ylabel("Accuracy")

plt.legend()

plt.show()
```



# CONCLUSION

This project successfully demonstrates the application of machine learning and deep learning techniques for the early detection of heart disease. By analyzing clinical and lifestyle data, the system was able to predict heart disease risk with high accuracy and reliability. The implementation of multiple machine learning models provided a comprehensive comparison of different predictive approaches, while the deep learning model effectively captured complex non-linear relationships within the dataset.

Data preprocessing techniques such as handling missing values, feature scaling, and stratified sampling played a crucial role in improving model performance and ensuring unbiased evaluation. Exploratory data analysis helped in understanding key risk factors and their relationship with heart disease, providing meaningful insights for both model development and medical interpretation.

The user-interactive prediction module enhances the practical applicability of the system by allowing real-time risk assessment based on patient input. This feature simulates a real-world clinical decision support system and demonstrates the potential of artificial intelligence in assisting healthcare professionals.

Overall, the proposed heart disease detection system highlights the effectiveness of artificial intelligence in healthcare applications. The project emphasizes the importance of early diagnosis, data-driven decision-making, and preventive care, contributing to improved patient outcomes and efficient healthcare services.

# KEY OBSERVATION

1. The dataset shows a class imbalance, with a higher number of patients without heart disease compared to patients with heart disease, making stratified sampling essential for fair model evaluation.

2. Age was observed to be a significant risk factor, with heart disease cases increasing noticeably in higher age groups.

3. Patients with hypertension (high blood pressure) and diabetes exhibited a higher likelihood of developing heart disease compared to healthy individuals.

4. Smoking habits, especially the number of cigarettes smoked per day, showed a strong association with heart disease risk.

5. Higher levels of cholesterol and systolic blood pressure were commonly observed among patients diagnosed with heart disease.

6. Exploratory data analysis revealed meaningful correlations between clinical features such as BMI, blood pressure, glucose levels, and heart disease occurrence.

7. Among the machine learning models, Random Forest and Support Vector Machine demonstrated better performance compared to simpler models due to their ability to capture non-linear relationships.

8. The deep learning MLP model achieved the highest accuracy by learning complex patterns within the dataset.

9. Feature importance analysis highlighted age, blood pressure, cholesterol, smoking status, and diabetes as the most influential predictors.

10. The user-input-based prediction system successfully demonstrated real-time heart disease risk assessment, indicating the practical applicability of the proposed system

# FUTURE SCOPE

1. The system can be enhanced by integrating **real-time patient data** from hospitals and electronic health records for continuous monitoring.

2. Advanced deep learning architectures and **hybrid models** can be explored to further improve prediction accuracy and robustness.

3. The project can be extended to include **medical imaging data** such as ECG, X-ray, or MRI for more comprehensive heart disease analysis.

4. Deployment of the system as a **web or mobile application** can enable easy access for healthcare professionals and patients.

5. Explainable AI techniques can be incorporated to improve **model transparency and interpretability** for clinical decision-making.

6. The system can be integrated with **wearable devices** for real-time heart health monitoring and early alert generation.

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